

STAT 2020 Discussion Slides

Week 14: One-Sample and Two-Sample Z/t Tests for Means

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Unified Hypothesis Setup and Notation

Throughout this week, write hypotheses as:

$$H_0 : \theta = \theta_0 \quad \text{vs.} \quad H_a : \theta > \theta_0, \theta < \theta_0, \text{ or } \theta \neq \theta_0.$$

Common symbols:

- One-sample mean: $\theta = \mu$
- Two-sample mean difference: $\theta = \mu_1 - \mu_2$, usually $\theta_0 = d_0$ (often $d_0 = 0$)
- Test statistics: Z^* for Z-tests, t^* for t-tests
- One-sample t: $df = n - 1$; pooled two-sample t: $df = n_1 + n_2 - 2$

Reminder About Confidence Intervals

- Unless explicitly stated otherwise, **“the confidence interval” means the standard two-sided CI.**
- In one-sided tests, a corresponding one-sided CI can also be reported.
- In this slide set, each worked test includes a matching one-sided CI to align with the tail direction.

Which Test to Use? (Quick Menu)

- One sample, σ known $\Rightarrow$ one-sample Z test
- Two independent samples, σ_1, σ_2 known $\Rightarrow$ two-sample Z test
- One sample, σ unknown $\Rightarrow$ one-sample t test
- Two independent samples, $\sigma_1 = \sigma_2$ unknown $\Rightarrow$ pooled two-sample t test

One-Sample Z: Formula Template

Test statistic:

$$Z^* = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}.$$

For right-tailed $H_a : \mu > \mu_0$:

Reject H_0 if $Z^* > z_\alpha$, $p\text{-value} = P(Z > Z^*)$.

Matching $(1 - \alpha)100\%$ lower one-sided CI:

$$\left(\bar{X} - z_\alpha \frac{\sigma}{\sqrt{n}}, \infty \right).$$

Example 1

Example 1: One-Sample Z (Right-Tailed)

Data: $n = 64$, $\bar{X} = 9.3$, $\sigma = 2.5$. Test at $\alpha = 0.05$:

$$H_0 : \mu = 8.5 \quad \text{vs.} \quad H_a : \mu > 8.5.$$

Step 1: Setup above.

Step 2:

$$Z^* = \frac{9.3 - 8.5}{2.5/\sqrt{64}} = 2.56, \quad z_{0.05} = 1.645.$$

Since $2.56 > 1.645$, reject H_0 .

Step 3: Evidence supports $\mu > 8.5$.

p-value: $P(Z > 2.56) = 0.0052$.

Matching 95% lower one-sided CI for μ :

$$\left(9.3 - 1.645 \cdot \frac{2.5}{8}, \infty \right) = (8.79, \infty).$$

Two-Sample Z: Formula Template

Use when two samples are independent and σ_1, σ_2 are known.

$$Z^* = \frac{(\bar{X}_1 - \bar{X}_2) - d_0}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}.$$

For right-tailed $H_a : \mu_1 - \mu_2 > d_0$:

Reject H_0 if $Z^* > z_\alpha$, $p\text{-value} = P(Z > Z^*)$.

Matching $(1 - \alpha)100\%$ lower one-sided CI for $\mu_1 - \mu_2$:

$$\left((\bar{X}_1 - \bar{X}_2) - z_\alpha \sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}, \infty \right).$$

Example 2

Example 2: Two-Sample Z (Right-Tailed)

Data: $\bar{X}_1 = 52.1$, $\bar{X}_2 = 50.8$, $n_1 = 36$, $n_2 = 64$, $\sigma_1 = 3$, $\sigma_2 = 4$, $\alpha = 0.05$.

$$H_0 : \mu_1 - \mu_2 = 0 \quad \text{vs.} \quad H_a : \mu_1 - \mu_2 > 0.$$

Step 1: Setup above.

Step 2:

$$Z^* = \frac{52.1 - 50.8}{\sqrt{3^2/36 + 4^2/64}} = \frac{1.3}{0.707} = 1.84, \quad z_{0.05} = 1.645.$$

Since $1.84 > 1.645$, reject H_0 .

Step 3: Evidence supports $\mu_1 > \mu_2$.

p-value: $P(Z > 1.84) = 0.0329$.

Matching 95% lower one-sided CI for $\mu_1 - \mu_2$:

$$(1.3 - 1.645 \cdot 0.707, \infty) = (0.14, \infty).$$

One-Sample t: Formula Template

Use when one sample and σ is unknown.

$$t^* = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}, \quad T \sim t_{n-1} \text{ under } H_0.$$

For left-tailed $H_a : \mu < \mu_0$:

$$\text{Reject } H_0 \text{ if } t^* < -t_{\alpha, n-1}, \quad p\text{-value} = P(T_{n-1} < t^*).$$

Matching $(1 - \alpha)100\%$ upper one-sided CI for μ :

$$\left(-\infty, \bar{X} + t_{\alpha, n-1} \frac{s}{\sqrt{n}} \right).$$

Example 3

Example 3: One-Sample t (Left-Tailed)

Data: $n = 16$, $\bar{X} = 48.8$, $s = 3.2$. Test at $\alpha = 0.05$:

$$H_0 : \mu = 50 \quad \text{vs.} \quad H_a : \mu < 50.$$

Step 1: Setup above.

Step 2:

$$t^* = \frac{48.8 - 50}{3.2/\sqrt{16}} = -1.50, \quad \text{df} = 15, \quad -t_{0.05,15} = -1.753.$$

Since $-1.50 > -1.753$, fail to reject H_0 .

Step 3: Not enough evidence that $\mu < 50$.

p-value: $P(T_{15} < -1.50) \approx 0.077$.

Matching 95% upper one-sided CI for μ :

$$\left(-\infty, 48.8 + 1.753 \cdot \frac{3.2}{4} \right) = (-\infty, 50.20).$$

Two-Sample Pooled t: Formula Template

Use when samples are independent and population variances are assumed equal.

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}},$$

$$t^* = \frac{(\bar{X}_1 - \bar{X}_2) - d_0}{s_p \sqrt{1/n_1 + 1/n_2}}, \quad T \sim t_{n_1+n_2-2} \text{ under } H_0.$$

For right-tailed $H_a : \mu_1 - \mu_2 > d_0$:

$$\text{Reject } H_0 \text{ if } t^* > t_{\alpha, n_1+n_2-2}, \quad p\text{-value} = P(T_{n_1+n_2-2} > t^*).$$

Example 4

Example 4: Two-Sample Pooled t (Right-Tailed)

Data: $\bar{X}_1 = 81.5$, $\bar{X}_2 = 78.6$, $n_1 = 12$, $n_2 = 10$, $s_1 = 4.0$, $s_2 = 3.5$, $\alpha = 0.05$.

$$H_0 : \mu_1 - \mu_2 = 0 \quad \text{vs.} \quad H_a : \mu_1 - \mu_2 > 0.$$

Step 1: Setup above.

Step 2:

$$s_p = \sqrt{\frac{11(4.0^2) + 9(3.5^2)}{20}} = 3.78, \quad t^* = \frac{81.5 - 78.6}{3.78\sqrt{1/12 + 1/10}} = 1.79.$$

$$\text{df} = 20, \quad t_{0.05, 20} = 1.725.$$

Since $1.79 > 1.725$, reject H_0 .

Step 3: Evidence supports $\mu_1 > \mu_2$.

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Example 4

Example 4 (Cont.): p-Value and CI

From $t^* = 1.79$ with $df = 20$:

$$p\text{-value} = P(T_{20} > 1.79) \approx 0.044.$$

Since $p < 0.05$, reject H_0 .

Matching 95% lower one-sided CI for $\mu_1 - \mu_2$:

$$(2.9 - 1.725 \cdot 1.62, \infty) = (0.10, \infty).$$

Take-Home Messages

- Always decide test type first (one/two sample, Z/t, known/unknown variance).
- Keep notation consistent: Z^* , t^* , z_α , $t_{\alpha,\nu}$, d_0 , s_p .
- Use the same 3-step structure every time.
- p-value method and critical-value method should agree.
- Unless specified, “the confidence interval” means the two-sided CI.

In-Class Attendance Check (Simple)

A random sample has $n = 25$, $\bar{X} = 10.6$, $s = 2.5$.

Test at $\alpha = 0.05$:

$$H_0 : \mu = 10 \quad \text{vs.} \quad H_a : \mu > 10.$$

At $\alpha = 0.05$, perform the appropriate test and report: test type, test statistic, p-value, and decision.

Please submit: Full Name; Section Number; Your Answer

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Answer (concise):

One-sample t test (σ unknown): $t^* = 1.20$, $p = P(T_{24} > 1.20) \approx 0.121 > 0.05$, so fail to reject H_0 .